Simulation-Based Inference and Maximum Likelihood Estimation

1. Introduction

This project applies core statistical inference techniques to both simulated and real-world datasets. The analysis is divided into two main components:

- **Section 1:** Simulation-based evaluation of estimators under clean and contaminated data conditions
- **Section 2:** Logistic regression modelling using Maximum Likelihood Estimation (MLE) to analyse the relationship between economic conditions and health outcomes

The primary objective is to demonstrate how statistical methods can be used to:

- Evaluate estimator performance using Monte Carlo simulation
- Understand the impact of skewness and outliers
- Apply likelihood-based modelling techniques
- Quantify uncertainty using statistical inference

This project is highly relevant to data science, as it combines **simulation, modelling, inference, and real-world interpretation**.

2. Section 1: Simulation-Based Inference

2.1 Problem Context

The goal is to estimate a "typical" completion time for employee security training. Completion times are assumed to follow a **right-skewed exponential distribution**, which is common in real-world duration data.

Three estimators are evaluated:

- Sample Mean
- Sample Median
- 10% Trimmed Mean

Their performance is assessed under two scenarios:

1. **Clean data** (ideal conditions)
2. **Contaminated data** (presence of outliers due to logging errors)

2.2 Methodology

A **Monte Carlo simulation** approach was used:

- Number of simulations: **1000**
- Sample size per simulation: **n = 250**
- Distribution:
  - Clean data: Exponential distribution (λ = 1)
  - Contaminated data:
    - 90% from Exp(1)
    - 10% from Exp(0.02) (heavy-tailed → introduces outliers)

2.3 Performance Metrics

Each estimator was evaluated using:

- **Bias**
  Difference between estimated and true value
- **Monte Carlo Standard Deviation (MCSD)**
  Measures variability across simulations
- **Mean Squared Error (MSE)**
  Combines bias and variance into a single measure

2.4 Results: Clean Data

Estimator	Bias	MCSD	MSE
Mean	0.0001	0.0621	0.0038
Median	-0.3050	0.0608	0.0967
Trimmed Mean	-0.1679	0.0550	0.0312

Interpretation

- The **mean is unbiased** and achieves the lowest MSE, making it the most efficient estimator under ideal conditions.
- The **median is negatively biased** due to the skewness of the exponential distribution.
- The **trimmed mean provides a balance**, with reduced variability but some bias.

Key Insight

When model assumptions are satisfied, the mean is the **optimal estimator** due to its efficiency.

2.5 Results: Contaminated Data

Estimator	Bias	MCSD	MSE
Mean	4.9373	1.3813	26.2825
Median	-0.1871	0.0821	0.0417
Trimmed Mean	0.1163	0.1686	0.0419

2.6 Interpretation of Contamination Effects

- The **mean becomes highly unstable**, with a dramatic increase in bias and MSE.
- The **median remains robust**, showing minimal change in performance.
- The **trimmed mean performs similarly to the median**, offering a compromise between efficiency and robustness.

Key Insight

Outliers have a **disproportionate impact on the mean**, but limited effect on robust estimators such as the median and trimmed mean.

2.7 Estimator Comparison and Recommendation

- **Mean**
  - Best under clean data
  - Performs poorly with outliers
- **Median**
  - Most robust estimator
  - Stable under contamination
- **Trimmed Mean**
  - Balanced performance
  - Good alternative when moderate robustness is needed

Recommendation

For real-world applications involving skewed or noisy data, the **median is the most reliable estimator**, while the trimmed mean provides a practical compromise.

2.8 Practical Implications

- In real systems (e.g., employee training tracking), data may contain errors or delays.
- Using the mean may lead to **misleading conclusions** due to extreme values.
- Robust measures such as the median ensure **more accurate and fair reporting**.

3. Section 2: Logistic Regression & Maximum Likelihood Estimation

3.1 Problem Context

This section models the probability that a country's life expectancy exceeds the global median using:

- Predictor: **log(GDP per capita)**
- Outcome: Binary variable (above/below median life expectancy)

This analysis connects statistical modelling with **real-world global health outcomes**, aligning with **UN Sustainable Development Goal 3 (Good Health and Well-being)**.

3.2 Model Specification

The logistic regression model is defined as:

$$P(Y = 1) = \frac{1}{1 + \exp\left(-(\beta_0 + \beta_1 x)\right)}$$

Where:

- $Y = 1$: Life expectancy above median
- $x = \log$ (GDP per capita)

3.3 Parameter Estimation (MLE)

Parameters were estimated using:

- **Manual MLE via optimisation (optim)**
- **Validation using GLM (glm)**

Estimated Coefficients

Parameter	Estimate
β_0	-20.382

Parameter	Estimate
β_1	2.349

3.4 Interpretation of Results

- The positive coefficient (**$\beta_1 > 0$**) indicates that higher GDP is associated with higher life expectancy.
- The relationship is **strong and statistically significant**.
- The magnitude of β_1 suggests a substantial increase in probability as GDP rises.

3.5 Fisher Information and Uncertainty

The **Hessian matrix** was used to compute the Fisher Information, allowing estimation of:

- Variance-covariance matrix
- Standard errors

Variance-Covariance Matrix

	β_0	β_1
β_0	11.457	-1.303
β_1	-1.303	0.149

3.6 Confidence Intervals

Parameter	Lower	Estimate	Upper
β_0	-27.017	-20.382	-13.748
β_1	1.592	2.349	3.106

Interpretation

- Both intervals exclude zero → strong statistical significance
- The slope coefficient remains strictly positive → confirms relationship

3.7 Model Stability (Profile Likelihood Insight)

The profile likelihood analysis shows:

- A clear peak at the estimated value of β_1
- Strong curvature indicating precise estimation
- Stable parameter behaviour

This confirms that the model is **well-identified and reliable**.

3.8 Real-World Interpretation

- A 10% increase in GDP corresponds to a meaningful increase in the probability of higher life expectancy
- Doubling GDP significantly increases the likelihood of improved health outcomes

This demonstrates a strong relationship between **economic development and population health**.

3.9 Policy Implications

The findings support investment in:

- Healthcare systems
- Education

- Infrastructure
- Public health programs

These factors are closely associated with higher GDP and improved life expectancy.

4. Conclusion

This project demonstrates the application of statistical inference techniques to both simulated and real-world data.

Key conclusions include:

- The **mean is efficient under ideal conditions**, but highly sensitive to outliers
- The **median and trimmed mean provide robustness in real-world datasets**
- Logistic regression using MLE effectively models complex relationships
- GDP is a **strong predictor of life expectancy**, with statistically significant effects

Overall, the project highlights the importance of:

- Choosing appropriate estimators
- Understanding data characteristics
- Applying statistical models carefully
- Interpreting results in real-world context